

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
Glenn Research Center (GRC)
JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
(JOFOC)
For 20 Watt 20 Kelvin Cryocooler for Thermal Control of Space-Based Liquid
Hydrogen NNC14CA15C

1. **Federal Acquisition Regulation (FAR) 6.303-2(b)(1) – Identification of the agency and the contracting activity, and specific identification of the document as a “Justification for other than full and open competition.”**

This document is a justification for other than full and open competition prepared by National Aeronautics and Space Administration (NASA) Glenn Research Center (GRC).

The procuring agency is the National Aeronautics and Space Administration (NASA) and the contracting activity is GRC.

2. **FAR 6.303-2(b)(2) – The nature and/or description of the action being approved:**

This justification provides the rationale for contracting by other than full and open competition to award a new work modification to contract number NNC14CA15C with Creare, LLC. This would add the requirement of Creare performing a vibration and post-vibration verification testing before shipment of the cryocooler to NASA.

3. **FAR 6.303-2(b)(3) – A description of the supplies or services required, to meet the Agency’s needs (including the estimated value):**

The services Creare is to perform include a vibration test of the integrated system and post-vibration cryocooler thermodynamic testing at 20 Kelvin before shipment of the 20W 20K Cryocooler to NASA. The agency requires these services because vibration testing is a required environment test to raise the technology readiness level to 6. In order to perform a vibration test, a vibration test fixture is required to be designed and fabricated, providing structural support to the cryocooler components during the test. The total estimated cost will be \$800,000.00. The addition of this work is also expected to add at least three (3) months to the current period of performance.

4. **FAR 6.303-2(b)(4) – An identification of the statutory authority permitting other than full and open competition:**

The statutory authority permitting other than full and open competition is 10 U.S.C. 3204(a)(1), as implemented by FAR 6.302-1 Only one responsible source and no other supplies or services will satisfy agency requirements.

5. FAR 6.303-2(b)(5) – A demonstration that the proposed contractor’s unique qualifications or the nature of the acquisition requires use of the authority cited:

The rationale supporting the use of 10 U.S.C. 3204(a)(1) is the services needed by the agency are available only from one responsible source services will satisfy the needs of the agency as described below:

This procurement is for the additional vibration testing of the 20W 20K cryocooler designed and built by Creare as part of a SBIR Phase III contract. As part of this vibration testing, a vibration test fixture will need to be designed and fabricated to provide structural support to the cryocooler components (three compressors, five recuperators, one turboalternator, tubing, valves). Many of these components contain proprietary design features that will be sensitive to vibrational loads and require in-depth analysis. Creare has the unique qualification of being familiar with the design of the system and having already performed vibration testing on the design for each of the individual components.

6. FAR 6.303-2(b)(6) – A description of the efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by Subpart 5.2 and, if not, which exception under 5.202 applies:

A notice to the Government Point of Entry (GPE) website (Sam.gov) was published on October 10, 2024, and closed on October 25, 2024, in accordance with FAR Subpart 5.2. This posting informed potential sources of NASA’s intent to award this sole-source modification to Creare. The results of this synopsis are summarized in Section 10 below.

The same notice to the Government Point of Entry (GPE) website (Sam.gov) was published on December 6, 2023, and closed on December 21, 2023, in accordance with FAR Subpart 5.2. This posting informed potential sources of NASA’s intent to award this sole-source modification to Creare. The results of this synopsis are also summarized in Section 10 below.

7. FAR 6.303-2(b)(7) – A determination by the contracting officer that the anticipated cost to the Government will be fair and reasonable:

The Contracting Officer’s signature on this document indicates that the Contracting Officer has determined that the anticipated cost to the government will be fair and reasonable. The Contractor shall be required to submit a proposal to be evaluated and negotiated by the Government. Prior to execution of the contractual instrument, a proposal analysis will be performed. The proposal analysis will ensure that the final agreed-to price for the contract action is fair and reasonable. The guidance and techniques specified at FAR 15.404 and NFS 1815.404 will be followed during the proposal evaluation. If appropriate, price analysis and technical analysis per FAR 15.404-1(b) and (e), and NFS 1815.404-1(e) will be performed to determine and negotiate a fair and reasonable price. If price analysis is not appropriate, then cost analysis and technical analysis, per FAR 15.404-1(c) and (e), and NFS 1815.404-1(e) will be performed to determine and negotiate a fair and reasonable price.

8. FAR 6.303-2(b)(8) – Description of the market research conducted, and the results, or a statement of the reasons market research was not conducted (e.g., urgent, and compelling requirement):

Prior to releasing the Sole Source Synopsis, Market Research was conducted by the NASA GRC technical team to investigate whether there were other potential contractors in the marketplace that could satisfy the NASA GRC anticipated project needs. This market research included personal knowledge in the procurement of the design/fabrication of a test fixture and the vibration testing, contact with knowledgeable people in industry regarding the commercial nature of this requirement and standard industry practices in this area of supply/service. Based upon the initial information presented by the technical team, the NASA GRC procurement team was informed that the technical team believed that only one company would be able to meet the Government's specific needs.

In order to verify this information, a Notice of Intent to Sole Source was posted via SAM.gov and there were no other contractors who responded stating they could meet the Government's specific need.

The Contracting Officer also searched government databases but did not find any other vendor who could meet all of the government's requirements.

Thus, Creare is the only contractor who can meet the government's requirement.

9. FAR 6.303-2(b)(9) – Any other facts supporting the use of other than full and open competition:

None.

10. FAR 6.303-2(b)(10) – A listing of the sources, if any, that expressed an interest in writing in the acquisition:

A notice of NASA's intent to award this sole-source action was synopsisized on the GPE website (Sam.gov) per FAR Subpart 5.2 (See Section 6 above). The notice was published on October 10, 2024 for the fifteen (15) day posting requirement, and industry was welcomed to challenge the sole source approach until October 25, 2024. That posting can be found at <https://sam.gov/opp/5b81eff1a2ef47c290ad3d243b9beb1f/view>.

This posting informed potential sources of NASA's intent to award this sole-source; the Contracting Officer did not receive any challenges of the Government's intent to sole-source the requirement herein.

This notice of intent to sole source was posted ten months earlier, on December 6, 2023 for the fifteen (15) day posting requirement, and industry was welcomed to challenge the sole source approach until December 21, 2023. That posting can be found at <https://sam.gov/opp/b0495b638073426a95fd6ba5a7258e21/view>. The government did not receive any challenges from industry of the Government's intent to sole-source the

requirement. A full JOFOC was completed however, the project decided to hold off on pursuing the vibration test, as a result of project priorities.

11. FAR 6.303-2(b)(11) – A statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required:

At this time, the marketplace for the above-mentioned procurement is virtually non-existent. However, if a future requirement does reoccur, a procurement approach will be used to re-examine the marketplace to the greatest extent possible in order to seek competition to the greatest extent possible.

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Hydrogen NNC14CA15C

JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION
SIGNATURE PAGE

Use the following signature authority for proposed actions over \$750K, but not exceeding \$15M:

Technical Representative: I certify that the facts presented in this justification are accurate and complete.

Benjamin
Nugent

Digitally signed by
Benjamin Nugent
Date: 2024.11.13
12:56:31 -05'00'

Benjamin Nugent
Technical Representative

Contracting Officer: I hereby certify that the above justification is complete and accurate to the best of my knowledge and belief.

Linda
Nabors

Digitally signed by
Linda Nabors
Date: 2024.11.13
16:36:01 -05'00'

Linda Nabors
Contracting Officer

APPROVED:

LEAHMARIE
KOURY

Digitally signed by
LEAHMARIE KOURY
Date: 2024.11.14
13:40:04 -05'00'

Leahmarie Koury
Competition Advocate
Glenn Research Center